Back to Basics: Raw Curvature Rivals HAWQ-V2's Product for Identifying Quantization-Damaged Layers

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Abstract

Hessian-based curvature analyses are a tool for quantization-sensitivity ranking, typically combining curvature and perturbation magnitude into a single product, $\mathrm{Tr}(H) \cdot \|\Delta W\|^2$, to guide mixed-precision assignment. Whether it actually tracks per-layer loss damage under a specific, hardware-attractive quantization scheme is a live and practically consequential question we test directly. We isolate each layer's own weight-quantization damage by quantizing exactly that layer's weights under Power-of-Two (PoT) encoding and measuring the resulting increase in validation loss, across three CNN architectures (a compact CNN, ResNet-18, ResNet-50) on two datasets (ImageNet100, CIFAR10; post-training quantization), then test three predictors against it: raw curvature $\mathrm{Tr}(H)$, the perturbation term $\|\Delta W\|^2$ alone, and their HAWQ-V2 product. No single predictor is universally reliable: raw curvature identifies the true single most-damaged layer at rank 1 in 3 of 6 architecture$\times$dataset combinations, and its top-5 overlap with the five truly most-damaged layers is never worse than the other two predictors'. Neither the perturbation term nor the HAWQ-V2 product ever wins outright either. Which specific layer is most damaged is itself dataset- and architecture-dependent. The network's stem convolution dominates on ImageNet100, but not always on CIFAR10, where damage can be small and diffuse. Full-ranking Spearman correlation is weak overall, with a few exceptions. Robust top-k identification, not full-distribution correlation, is the finding that survives across datasets.

1 Introduction

Efficient inference on resource-constrained hardware makes weight quantization a central problem in deploying neural networks. Power-of-Two (PoT) quantization is a particularly hardware-attractive scheme because PoT weights are exact powers of two, multiply-accumulate operations reduce to bitshifts, avoiding multipliers entirely [Miyashita et al., 2016, Przewłocka-Rus et al., 2022]. This efficiency comes at a real cost under naive Post-Training-Quantization, one that varies sharply by architecture and dataset and concentrates unevenly across layers.

The Hessian-Aware-Quantization (HAWQ) family addresses this by using curvature as a sensitivity signal for mixed-precision decisions. HAWQ [Dong et al., 2019] ranks layers by the top Hessian eigenvalue; HAWQ-V2 [Dong et al., 2020] refines this into a curvature-times-perturbation product, $\mathrm{Tr}(H) \cdot \|\Delta W\|^2$, motivated by a second-order Taylor expansion of the loss. Per-layer traces are estimated via Hutchinson's method implemented in PyHessian [Yao et al., 2020].

HAWQ uses this formulation to decide bit-width allocation. This claim – that $\mathrm{Tr}(H) \cdot \|\Delta W\|^2$ tracks per-layer damage – was recently tested in the LLM setting [Hill, 2026] but remains untested under

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PoT for vision CNNs; neither HAWQ [Dong et al., 2019] nor HAWQ-V2 [Dong et al., 2020] validates it per-layer, relying instead on end-to-end accuracy after allocation. PoT's non-uniform grid ties perturbation size to grid alignment, not layer size or precision, leaving the product's predictive power here untested.

Since HAWQ-V2's product is itself motivated by a second-order Taylor expansion of the loss (§5), we take per-layer isolated loss damage as the direct target this motivation implies (§3.2). We test three predictors, raw curvature $S_{raw} = \mathrm{Tr}(H)$, perturbation magnitude $S_{pert} = \|\Delta W\|^2$, and the HAWQ-V2 product $S_{hawq} = \mathrm{Tr}(H) \cdot \|\Delta W\|^2$, against per-layer weight-quantization loss damage on three CNN architectures and two datasets. None of the predictors we used is universally reliable. Raw curvature identifies the true top-damage layer at rank 1 in half of the six combinations, and never has worse top-5 overlap with true damage than the other two predictors, which never win outright either; which layer is most damaged, and how well any predictor finds it, depends on the dataset and architecture and on how concentrated total damage is – diffuse low-damage regimes (ResNet-50/CIFAR10) defeat all three predictors; and careful canonical trace measurement matters.

Our scope is weight quantization: isolated weight-only damage matches or exceeds total joint damage in every combination we test while isolated activation-only damage stays within ± 0.4 points, and activation quantization follows a different, outlier-driven failure mode [Xiao et al., 2023, Lin et al., 2024] that we do not address here.

2 Related Work

HAWQ [Dong et al., 2019] uses the top Hessian eigenvalue to guide mixed-precision bit-width assignment while HAWQ-V2 [Dong et al., 2020] refines this to a curvature-times-perturbation sensitivity product motivated by a second-order Taylor expansion of the loss. Both use per-layer Hessian trace estimates obtained via Hutchinson's stochastic estimator, popularized as a standard toolkit by PyHessian [Yao et al., 2020], which we also use here.

PoT was proposed as a hardware-friendly logarithmic weight representation [Miyashita et al., 2016], further motivated for low-bitwidth hardware compliance [Przewłocka-Rus et al., 2022]. On the activation side, excluded from our scope, SmoothQuant [Xiao et al., 2023] and AWQ [Lin et al., 2024] show activation quantization fails via channel-wise outliers, not curvature; BRECQ [Li et al., 2021] addresses post-training quantization via block-wise reconstruction.

3 Methods

3.1 Trace measurement

We estimate per-layer Hessian trace with PyHessian's Hutchinson estimator [Yao et al., 2020] using Rademacher probes. To keep results reproducible under configuration choices that the literature usually leaves implicit, we fix a canonical configuration. Mean-reduced cross-entropy loss, a fixed validation batch, three probe seeds (42, 43, 44) averaged into a mean±std per layer, deterministic algorithms, and `eval()` mode. Probe budget is matched within each dataset but differs across them (CIFAR10: 80 images, 100 Hutchinson iterations; ImageNet100: 24 images, 30 iterations, reduced because per-iteration Hessian-vector-product cost is far higher at 224×224 than at 32×32). We report the fused-BatchNorm basis as canonical for cross-stage comparison, since PTQ and Quantization-Aware-Training (QAT) checkpoints exist only in fused form. FP32 models are traced in both unfused and fused bases so the fusion effect is never hidden. Because PTQ/QAT weights are quantized with a straight-through estimator, the Hessian evaluated on these models is the STE-relaxed loss curvature [Bengio et al., 2013], not the classical Hessian of an unquantized network.

3.2 Weight-only damage isolation

For each layer l we define $\mathrm{loss\_damage}(l) = \mathrm{loss}_{iso}(l) - \mathrm{loss}_{fp32}$, where $\mathrm{loss}_{iso}(l)$ is the mean validation cross-entropy loss (identical loss definition to §3.1's trace estimator) of a model in which layer l's weights are PoT-quantized while all other weights and all activations remain FP32. Positive $\mathrm{loss\_damage}(l)$ means loss increased, i.e. damage, matching the sign convention below so the

Table 1: Rank assigned by each predictor to the true top-damage layer (1 = best; computed from $\mathrm{abs\_loss\_damage}$, §3.2) and each predictor's overlap with the true top-5 damaged layers (out of 5) – how many of the true top-5 this predictor's own top-5 recovers, not a rank position. The final column reports S_{raw}'s overlap with the true top-k^* damaged layers under a size-normalized $k^* = \lceil 0.1n \rceil$ per architecture; see Appendix E for the normalization rationale and per-architecture k^* values. Per architecture×dataset combination, PTQ, loss-based damage.

Dataset	Model (n)	top-damage layer	S_{raw} rank	S_{raw} top5	S_{pert} top5	S_{hawq} top5	S_{raw} top-k^*
ImageNet100	CNN (6)	`conv1`	3	4/5	4/5	4/5	0/1
ImageNet100	ResNet-18 (21)	`conv1`	**1**	2/5	0/5	1/5	1/3
ImageNet100	ResNet-50 (54)	`conv1`	**1**	1/5	0/5	0/5	1/6
CIFAR10	CNN (6)	`conv1`	2	4/5	4/5	4/5	0/1
CIFAR10	ResNet-18 (21)	`layer1.1.conv1`	**1**	3/5	0/5	0/5	1/3
CIFAR10	ResNet-50 (54)	`fc`	40	2/5	0/5	1/5	2/6

two targets never need to be mentally flipped against each other. We use $\mathrm{abs\_loss\_damage}(l) = |\mathrm{loss\_damage}(l)|$ as the primary target for predictor comparison (§4).

3.3 Three predictors

We compare three per-layer sensitivity scores against $\mathrm{abs\_loss\_damage}(l)$ (§3.2): raw curvature $S_{raw}(l) = \mathrm{Tr}(H_l)$ (canonical fused basis, evaluated on the FP32 model, unnormalized Hutchinson trace sum, not divided by parameter count); perturbation magnitude $S_{pert}(l) = \|W_l - Q(W_l)\|^2$; and the HAWQ-V2 product $S_{hawq}(l) = \mathrm{Tr}(H_l) \cdot \|W_l - Q(W_l)\|^2$ [Dong et al., 2020]. S_{raw} and S_{pert} are exactly the two multiplicative factors S_{hawq} is built from, so testing all three distinguishes whether damage is driven by curvature, by perturbation size, or by their product; see Appendix D for the full justification, including the unnormalized-trace choice. We evaluate each via Spearman rank correlation, the rank each predictor assigns to the layer of true maximum damage, and overlap between each predictor's top-5 and the true top-5 damaged layers.

4 Results

4.1 Setup

We study three architectures, a compact CNN (6 quantizable layers), ResNet-18 (21 layers), and ResNet-50 (54 layers), on two datasets (CIFAR10, ImageNet100), quantized post-training (PTQ) with PoT weight encoding, six architecture×dataset combinations in total. All numbers below are for PTQ, loss-based damage (§3.2) unless stated otherwise.

4.2 Which layer is most damaged is dataset- and architecture-dependent

In every combination, one layer absorbs disproportionately more weight-quantization damage than the rest (Figure 1), but which layer this is depends on both the dataset and the architecture: Table 1 shows the top-damage layer differs across all three architectures on CIFAR10 alone, so dataset is not the only axis of variation. On ImageNet100 it is always the stem convolution `conv1`, with loss damage 0.145, 0.225, and 0.533 for ResNet-18, ResNet-50, and CNN respectively. On CIFAR10 it is `conv1` only for the CNN (loss damage 0.197); for the ResNets a different layer takes over, `layer1.1.conv1` for ResNet-18 (0.029) and the final fully-connected layer for ResNet-50 (0.017), reflecting CIFAR10's far smaller and more diffusely spread PTQ damage.

Isolated single-layer quantization can also help: ResNet-50/CIFAR10's `fc` layer – Table 1's nominal top-damage layer only because ranking is by absolute value – in fact has signed loss damage -0.017 (a $\sim$3% reduction against baseline loss 0.565), a sign the $|\mathrm{loss\_damage}|$ ranking hides.

Table 1 summarizes how well each predictor identifies this layer under loss-based damage. S_{raw} achieves rank-1 identification in 3 of 6 combinations, and its top-5 overlap is never worse than S_{pert}'s or S_{hawq}'s in any combination; tied in two, strictly better in four. S_{pert} never achieves rank-1 identification, and buries the true top-damage layer near the bottom of the ranking whenever that layer is a low-fan-in convolution (`conv1`, `layer1.1.conv1`. See Figure 1). S_{hawq} likewise never

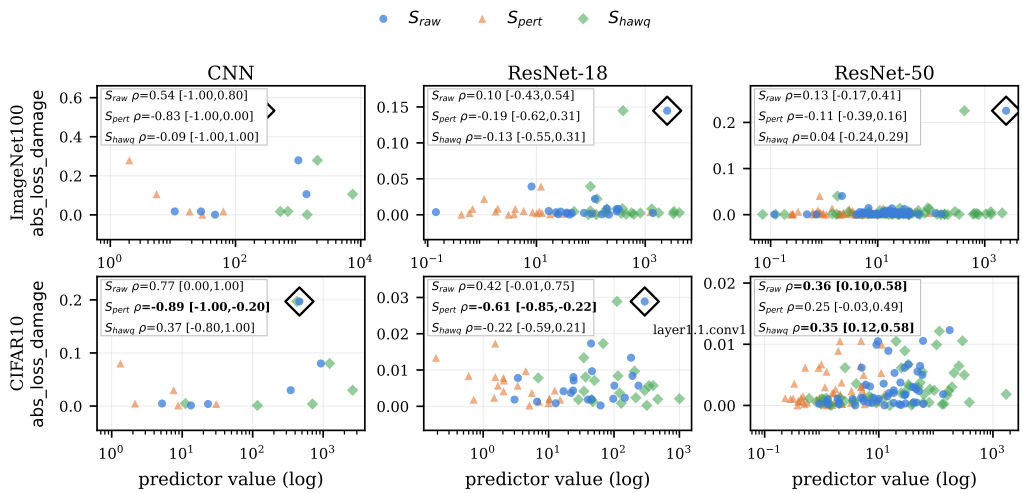

Figure 1: Per-layer predictor value (log scale) vs. measured abs_loss_damage, PTQ; top row ImageNet100, bottom row CIFAR10, columns left-to-right CNN/ResNet-18/ResNet-50. Open black diamonds mark each architecture's true top-damage layer (labeled where not `conv1`). Each panel's inset reports all three predictors' Spearman ρ with 95% bootstrap CI ($B = 2000$, resampling layers with replacement); bold denotes a CI excluding 0. S_{raw} (blue circles) tracks the top-damage layer well in 4 of 6 panels; S_{pert} (orange triangles) never does; S_{hawq} (green diamonds) scatters just as broadly as S_{pert} and likewise never concentrates near the top-damage layer, consistent with its top-5 overlap never exceeding S_{raw}'s (Table 1).

achieves rank-1 identification and never exceeds S_{raw}'s top-5 overlap. The size-normalized S_{raw} top-k^* column shows the CNN's apparent top-5 dominance is largely an artifact of only having 6 layers to choose from ($k^* = 1$ recovers nothing there), while the two ResNets ($k^* = 3$ and $k^* = 6$ respectively) retain a genuine, if partial, normalized-top-k signal. The one combination where all three predictors fail even at fixed top-5, ResNet-50/CIFAR10, is also the combination where damage is most diffusely spread across its 54 layers. We take this as evidence that a usable curvature signal requires damage concentrated enough to separate from noise, not a failure specific to any one predictor.

Full-ranking Spearman correlation (Figure 1's per-panel insets, 95% bootstrap CI, $B = 2000$) is fragile: 5 of 18 (predictor×architecture×dataset) correlations reach significance by the conventional $p < 0.05$ criterion (one fewer by CI excluding 0 alone, a boundary case at $n = 6$), and CIFAR10's smaller, more diffuse damage likely explains its weaker correlations relative to ImageNet100's. Top-k identification (Table 1), not full-distribution correlation, remains the robust cross-dataset summary.

5 Discussion and Limitations

HAWQ-V2's product is motivated by a second-order Taylor expansion of the loss, $\Delta L \approx \frac{1}{2}\Delta W^\top H \Delta W$, which assumes $\|\Delta W\|^2$ is comparable in scale across layers, so that curvature dominates the ranking. This is a reasonable approximation under uniform quantization, where perturbation size scales predictably with a layer's bit-width and weight range. Under PoT it breaks down: perturbation size depends on how well a layer's specific weight values happen to align with the PoT grid, which is close to incidental for a small, low-fan-in stem-like kernel. Consistent with this, S_{pert}'s loss-based correlation with abs_loss_damage (Figure 1) is negative in 5 of the 6 combinations, opposite in sign to S_{raw}'s positive correlation in every one of those 5 – pulling in different directions where the product (§3.3) is meant to add information.

More precisely, the product only behaves as sensitivity if $\|\Delta W\|^2$ is at least uncorrelated with a layer's curvature-based sensitivity. PoT structurally violates this, since grid alignment is a property of a layer's weight distribution, independent of its curvature, and for `conv1` the two are directly opposed (highest sensitivity, smallest perturbation). This is why an established, widely used sensitivity formulation misleads specifically under this quantization scheme, not a general failure of curvature-based sensitivity, and it holds regardless of which layer happens to be most damaged in a given architecture or dataset.

Limitations. One training run per architecture per dataset; cross-seed variance in the top-damage-layer finding is not characterized. We analyze PTQ only; QAT per-layer sensitivity, where isolated damage can also be negative, is left to future work. Our findings are specific to PoT quantization; applicability to uniform int8 or mixed-precision schemes is untested. See Appendix A for extended discussion of probe-budget effects and additional caveats.

6 Conclusion

No single curvature-based predictor is universally reliable for identifying quantization-damaged layers under PoT. While raw canonical Hessian trace identifies the true top-damage layer at rank 1 in half of the six architecture$\times$dataset combinations we test and never has worse top-5 overlap with true loss-based weight-quantization damage than the HAWQ-V2 product or the perturbation term alone, neither alternative ever wins outright either, across three CNN architectures on both ImageNet100 and CIFAR10 under PoT quantization. Which specific layer is most damaged is itself dataset- and architecture-dependent, and where total damage is small and diffuse (ResNet-50/CIFAR10), no predictor we test performs well. Robust top-k identification, not full-distribution correlation significance, is what survives across datasets and architectures. Curvature-based sensitivity formulations therefore need quantization-scheme- and damage-regime-specific adaptation rather than a single universal product. Future work includes a random-matrix-theory-informed decomposition of the bulk-versus-outlier spectrum, matching probe budgets across datasets to isolate estimation noise from genuine dataset effects, and systematic extension to uniform and mixed-precision quantization schemes.

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A Extended Limitations

Probe budgets differ between datasets (§3.1) and are not matched, so trace-estimation noise could itself contribute to CIFAR10's weaker correlations, independent of any genuine dataset effect.

We verified the reported findings against an accuracy-based version of the same analysis; rankings and predictor comparisons are qualitatively unchanged.

B Reconciliation of ResNet-50 `conv1` Trace Measurements

Canonical trace measurement matters more than the literature typically documents. Two historical, differently-configured runs in our pipeline produced ResNet-50 `conv1` PTQ trace values of 13161.8 and 1030.9 against our canonical re-measurement of 97.8 – ratios of 134.6× and 10.5×, a gap of two orders of magnitude for the same nominal quantity. We ran a systematic sweep over basis (fused/unfused), loss reduction, probe count, and data source specifically to reconcile these two historical values with the canonical measurement; neither reconciled within tolerance under any configuration we tried. We report this as a documented, closed finding, not an open question we simply have not gotten to, and recommend the same fixed-canonical-configuration discipline, reported explicitly, for future curvature-based quantization-sensitivity work.

The same sweep also touched two related historical anchors for the same layer, listed in Table 2 alongside the PTQ discrepancy above.

Table 2: Reconciliation ledger for ResNet-50 `conv1`: historical values recorded in earlier, differently-configured runs against the canonical re-measurement (§3.1). "Resolution" names the sweep configuration whose residual to the historical value was smallest; "unresolved" means no configuration in the sweep reproduced the historical value within tolerance. Elevation is $\mathrm{conv1\ trace}/\mathrm{median}$(all-layer trace), fused basis.

Quantity	Old value	Canonical value	Ratio	Resolution
`conv1` FP32 (run 1)	13.87	14.9	0.93×	unfused basis (residual 2.8%)
`conv1` FP32 (run 2)	11.79	14.9	0.79×	unresolved
`conv1` PTQ (run 1)	13161.8	97.8	134.6×	unresolved
`conv1` PTQ (run 2)	1030.9	97.8	10.5×	unresolved
`conv1` elevation, FP32	14.0	11.7	1.19×	unresolved

C Predictor Concordance Analysis

Table 3 reports pairwise Spearman correlation between the three predictors themselves (S_{raw}, S_{pert}, S_{hawq}), computed directly between predictor values and independent of which damage target (§3.2) is used, across all six architecture×dataset combinations.

Table 3: Pairwise Spearman ρ between predictor values across all six architecture×dataset combinations.

Model	Dataset	$\rho(S_{raw}, S_{pert})$	$\rho(S_{raw}, S_{hawq})$	$\rho(S_{pert}, S_{hawq})$
CNN	ImageNet100	−0.71	0.60	0.09
CNN	CIFAR10	−0.54	0.71	0.03
ResNet-18	ImageNet100	−0.60	0.43	0.41
ResNet-18	CIFAR10	−0.55	0.45	0.43
ResNet-50	ImageNet100	0.40	0.78	**0.83**
ResNet-50	CIFAR10	0.48	0.86	**0.84**

On ResNet-50, the product collapses toward its perturbation factor rather than behaving as an independent combination: the predictors' own pairwise concordance is $\rho = 0.84$ between S_{pert} and S_{hawq} on CIFAR10 and $\rho = 0.83$ on ImageNet100, so S_{hawq}'s failure to beat S_{raw} on this architecture is effectively S_{pert}'s failure carried through the product, not a genuine three-way test there.

We report this concordance analysis directly, rather than leaving it to future work, given its explanatory relevance to §5's finding that S_{hawq} never exceeds S_{raw}'s top-5 overlap on this architecture.

D Extended Justification for the Three Predictors

These three are not an arbitrary selection: S_{raw} and S_{pert} are exactly the two multiplicative factors HAWQ-V2's product (§5) is built from, isolating curvature and perturbation magnitude respectively with no information from the other. Testing all three tests HAWQ-V2's product form directly against each of its own constituent factors, rather than against unrelated alternatives, distinguishing whether per-layer damage is driven by curvature, by perturbation size, or genuinely by their product. Reporting the unnormalized trace, matching HAWQ-style usage, is a deliberate choice, not an oversight.

A parameter-normalized variant of S_{raw} (dividing by each layer's weight count) helps in 5 of the 6 combinations but actively hurts on ResNet-50/CIFAR10 (ρ drops from 0.36 to 0.16).

E Predictor Rank Normalization

The three architectures span an order of magnitude in layer count ($n = 6$, 21, 54), so a fixed top-5 threshold is nearly the whole network for the CNN but a small slice of ResNet-50 – not a comparable claim across architectures. Table 1's S_{raw} top-k^* column instead reports S_{raw}'s overlap with the true top-k^* damaged layers under a size-normalized $k^* = \lceil 0.1n \rceil$ (concretely 1, 3, and 6 layers for CNN, ResNet-18, and ResNet-50 respectively), so all three architectures are held to the same top-decile standard.